



B.I.G

A BST-Indexed Graph Model for Version Control

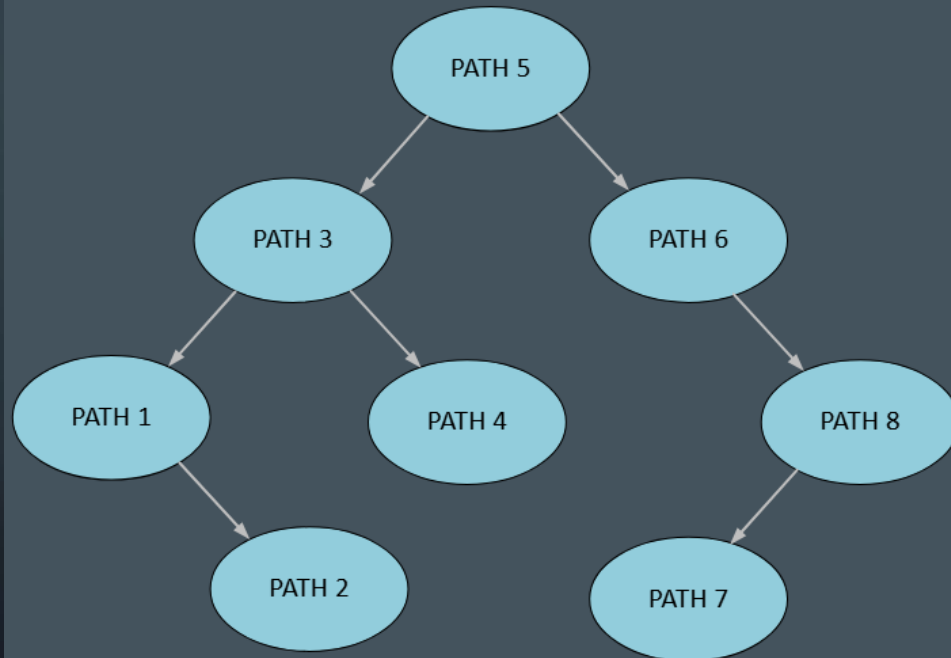
Final project in Data Structures course

Motivation

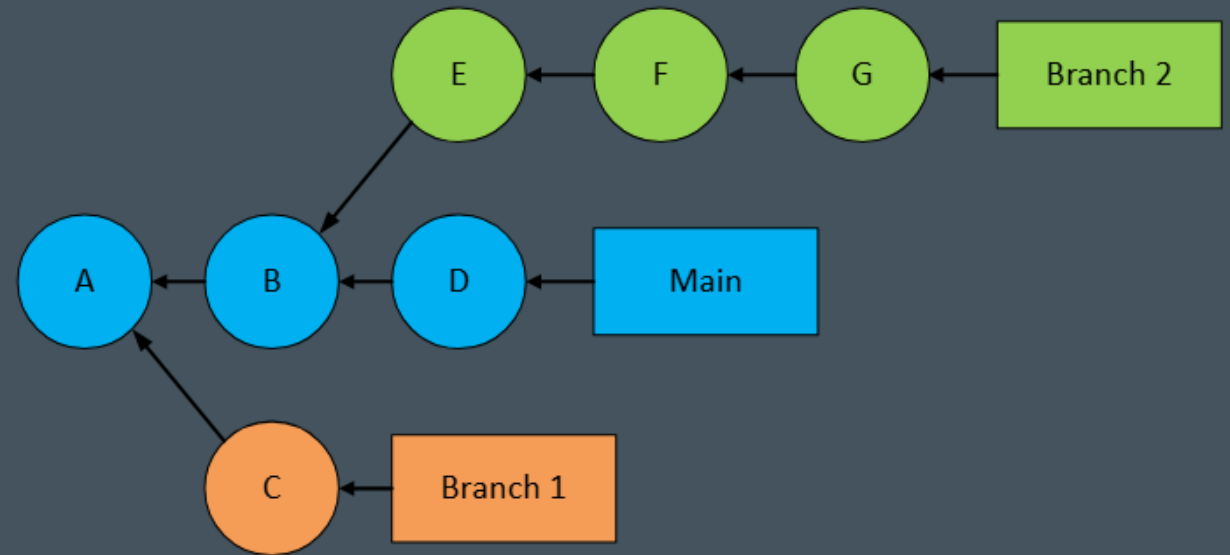


Methodology

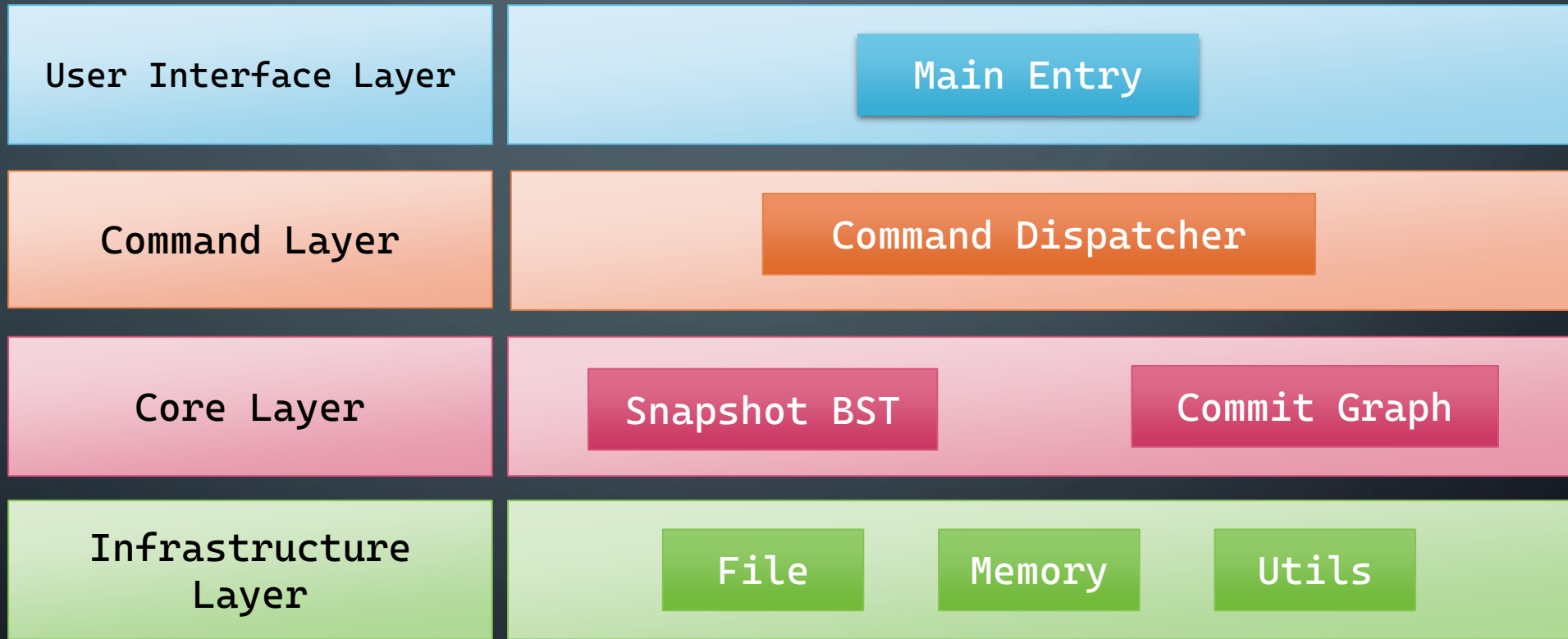
Binary Search Tree
(BST)



Directed Acyclic Graph
(DAG)



Methodology

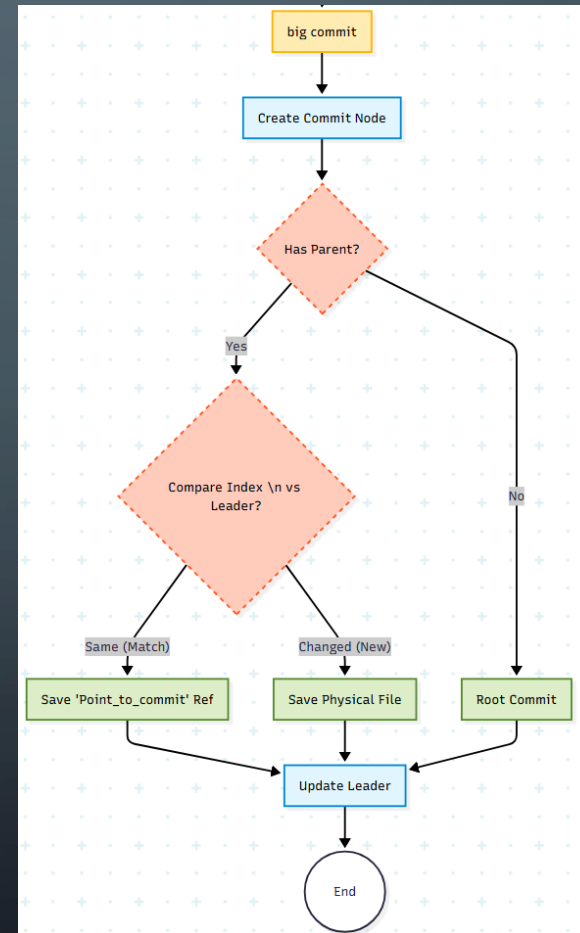
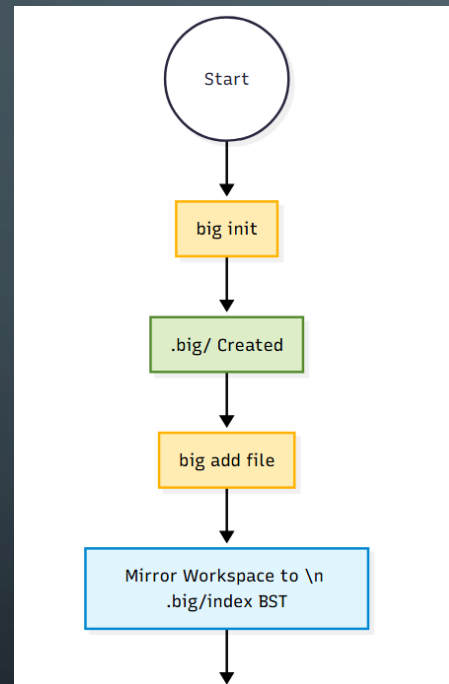


Development Environment

- **OS/Kernel:** Windows 11 with WSL2 (Ubuntu)
- **Language:** C (C17 Standard)
- **Compiler/Build:** GCC, Makefile
- **Tools:** VSCode, Valgrind, Git

Proposed Features – Phase 1 (The Core)

init -> add -> commit (The Core Workflow)

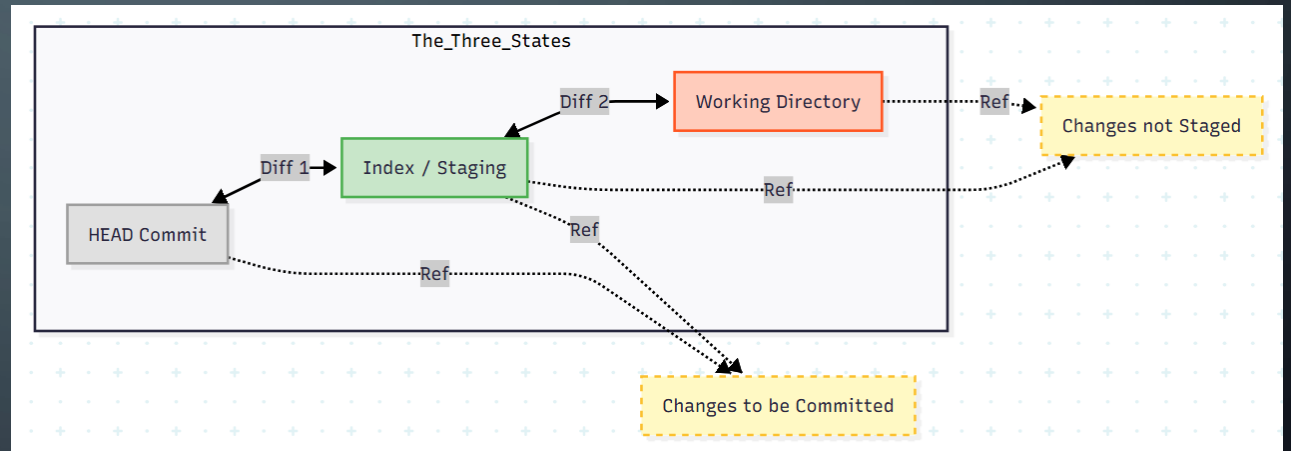


Phase 2 (Visibility)

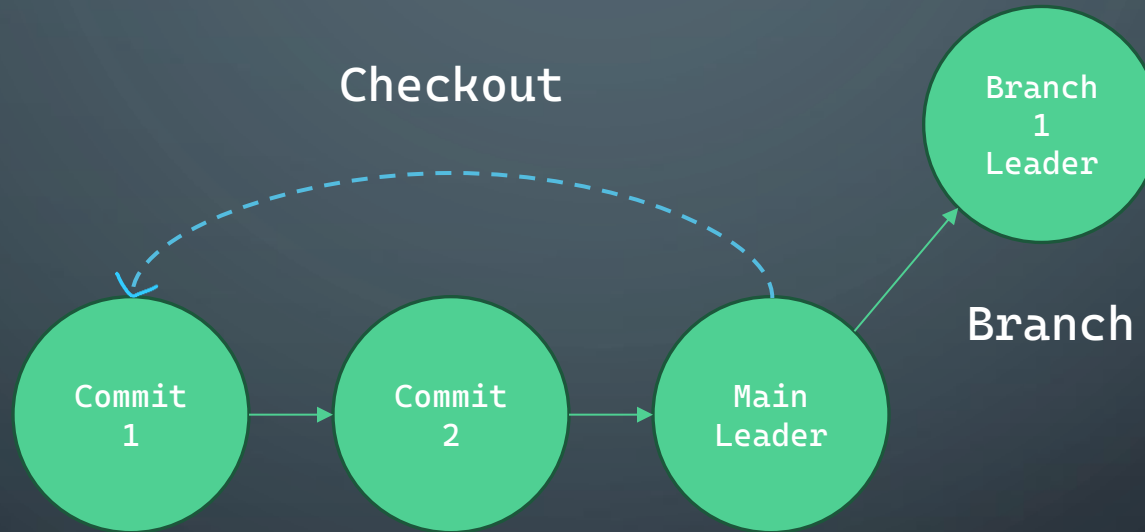
Log

```
306a8d5 feat: Implemnt 'big add' tracing commit path is content is same
048a3cb feat: 'big add' now can trace Leader and compare content
803db40 feat: Implement new 'big commit' with new 'big add' feature
2f10f74 docs: Add comments in main.c and utils/ code
3452e72 feat: Rewrite 'big add' feature creating a index directory include all files
0c90c65 feat: Implement 'big status' feature of tracing files are in index or not
189081a refactor: Replace malloc, free and fopen with xmalloc, xfree and xfopen
44d1f25 feat: Implement fopen wrapper
cf5ed60 feat: Implement malloc and free wrapper
986f2da build: More strict compiler check
```

Status



Phase 3 (Navigation)



Conclusion

Feature list:

- `init`
- `add`
- `commit`
- `log`
- `checkout`
- `branch`

What I will learn:

- System Programming (File I/O, Process Control).
- Memory Management (Manual Malloc/Free).
- Data Structures in Practice (BST + DAG).

The image features a dark blue background with a subtle gradient. In the corners, there are decorative white line art elements resembling circuit boards or neural networks, with lines and small circles. The text "THANK YOU FOR LISTENING." is centered in a white, bold, sans-serif font.

THANK YOU FOR LISTENING.